PUSHPARAJ ADKA

SENIOR EMBEDDED DEVELOPER

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Professional Summary

Accomplished Senior Embedded Systems Developer with 6+ years of experience designing and architecting high-efficiency firmware and IoT solutions. Adept at low-level driver optimization and protocol implementation while championing Agile methodologies to unify cross-functional teams. Recognized for transforming complex technical requirements into high-performance products and driving continuous process improvements.

Core Competencies

- **Architecture & OS:** FreeRTOS, ThreadX, Embedded Linux, Yocto Project, Device Tree configuration, Bootloaders (Serial/Ethernet)
- **Microcontrollers & Memory:** STM32N6, STM32H750, ATSAME54, ATmega, QSPI Memories, Hardware Bring-up
- **Protocols & Networking:** CAN FD, Ethernet, Modbus TCP, MQTT, TCP/IP (LwIP), UART, SPI, I2C, USB, AES70
- **Advanced Controls:** Motor Control Algorithms (FOC, 6-Step), BLDC Drivers (Hub/Drum), TouchGFX GUI Integration
- **Engineering Leadership:** Agile/Scrum Master, Team Mentorship, Cross-Functional Hardware/Software Alignment

Professional Experience

Embedded Controllers Lead | Bose Professionals Mangalore | 2025 - Present

- **Firmware Design & Bootloaders:** Spearheaded the architecture and development of embedded firmware for professional audio systems, designing secure serial/Ethernet bootloaders, persistent storage frameworks, and optimized low-level subsystem drivers (Ethernet, display, audio).
- **UI Integration & Protocols:** Architected robust communication protocols and software interfaces to bridge embedded controllers with UI applications; championed the TouchGFX display integration to deliver scalable device configuration and real-time monitoring.
- **Networked Audio Services:** Engineered advanced network applications and control services utilizing TCP/IP and AES70, successfully implementing automated device discovery, remote configuration, firmware management, and deterministic communication.
- **Embedded Linux & Kernel Integration:** Built and customized robust Embedded Linux platforms using the Yocto Project; executed low-level board bring-up, including Device Tree configurations, kernel driver integration, HDMI audio enablement, and USB functionality.

Senior Firmware Developer | Unbox Robotics | 2023 – 2025

- **Team Leadership & Driver Development:** Managed a team of 4 embedded systems engineers to design and implement low-level drivers (UART, timers, BLDC motors); utilized 6-step and Field-Oriented Control (FOC) algorithms to significantly enhance motor control precision and product functionality.

- **System Architecture & Mentorship:** Architected robust embedded systems and mentored junior developers, establishing coding standards and review workflows that ensured high-quality deliverables and efficient project timelines.
- **Agile Leadership (Scrum Master):** Served as Scrum Master for the BLDC driver development squad, driving Agile methodologies and cross-functional collaboration with software teams to engineer custom configuration tools, testing utilities, and optimized workflows.
- **AI-Driven Tooling & Lifecycle Management:** Developed innovative testing and quality control tools leveraging GPT models for automated PCB testing and driver validation, while managing the complete product lifecycle from initial concept to deployment.
- **Cross-Functional Optimization:** Spearheaded rigorous code reviews and partnered closely with controls, software, and hardware engineering departments to seamlessly integrate new features and maximize overall system performance.

Firmware Developer | Unbox Robotics | 2020 – 2023

- **Industrial IoT & Automation:** Implemented Modbus TCP and integrated MQTT over Ethernet for high-scale warehouse automation systems, significantly enhancing machine-to-machine (M2M) communication efficiency and network scalability.
- **AGV & Robotics Firmware:** Developed robust firmware for Autonomous Guided Vehicles (AGVs), custom microcontroller-based APIs, and smart charging systems utilizing the CAN protocol to optimize real-time driver control loops.
- **Device Management & OTA:** Architected and delivered seamless Over-the-Air (OTA) firmware update deployment mechanisms across Ethernet, CAN, and USB interfaces, ensuring zero-downtime and uninterrupted fleet operations.

Firmware Engineer | Dream Tech Innovation Pvt. Ltd. | Jan 2018 - Jul 2020

- **Product Development & Simplification:** Co-developed and optimized firmware for the DREAM KIT, creating an intuitive, abstraction-layered embedded platform designed to simplify complex electronics and coding concepts for students.
- **Full-Lifecycle Engineering:** Collaborated closely with cross-functional design teams from initial concept to commercial launch, validating system architecture to ensure high product reliability, functionality, and manufacturability.
- **Validation & Lifecycle Support:** Headquartered firmware testing, debugging, and verification protocols across pre-launch, manufacturing, and post-launch deployment phases to ensure field stability.

Embedded System Intern | Centre for Nano Science and Engineering (CeNSE) | 2017

- **Hardware-Software Co-Design:** Gained critical, hands-on experience bridging hardware and firmware domains, executing schematic analysis, signal routing, and rapid prototyping for embedded systems.
- **Firmware Development & Logic Design:** Designed and developed an automated channel detector utilizing hardware multiplexers and Arduino microcontrollers, optimizing signal acquisition and data throughput efficiency.

Robotic Process Automation | Chennai Institute of Technology | Jan 2018 - Jan 2018

- Programmed and deployed industrial robotic systems, optimizing trajectory planning and automation sequences for real-world manufacturing environments.

Key Projects

Professional Networked Audio & Display System *(Bose Professional)*

- Tech Stack: STM32N6, Embedded Linux (Yocto), TouchGFX, TCP/IP, AES70 protocol, QSPI Flash

- Impact: Architected core firmware for cloud-managed audio systems. Customized an Embedded Linux platform (Yocto/Device Tree) to enable HDMI audio/USB. Led a TouchGFX GUI integration over an STM32N6 MCU and deployed AES70 services for real-time device discovery and remote configuration.

Industrial AGV Fleet Platform *(Unbox Robotics)*

- Tech Stack: FreeRTOS, STM32H750, ATSAME54, CAN FD, Modbus TCP, MQTT, LiDAR
- Impact: Engineered real-time firmware for warehouse automation AGVs. Developed RTOS drivers to integrate motor controllers, actuators, and LiDAR arrays into driver control loops. Implemented Modbus TCP and MQTT over Ethernet pipelines to optimize fleet telemetry and M2M throughput.

Advanced BLDC Motor Driver Subsystem *(Unbox Robotics)*

- Tech Stack: Embedded C, Field-Oriented Control (FOC), 6-Step Algorithms, Python, GPT Scripting
- Impact: Led a 4-engineer squad designing firmware for hub/drum BLDC motors using FOC and 6-step algorithms to maximize torque efficiency. Developed automated, GPT-driven hardware validation and PCB diagnostic tools to streamline manufacturing QC.

Connected Barricading & Sorting Systems *(Unbox Robotics)*

- Tech Stack: MQTT over Ethernet, Sensor Arrays, Digital Actuators, Event-Driven Architecture
- Impact: Designed and deployed large-scale automated sorting safety infrastructure. Implemented an event-driven MQTT architecture to synchronize automated doors and parcel-detection tower lamps based on real-time sensor array polling.

Education

- **Bachelor's Degree in Electronics and Communication Engineering**

Visvesvaraya Technological University | *2016 – 2020*

- **Pre-University Education**

Parijnan Pre-University College | *2014 – 2016*

Certifications

- Microcontrollers and the C Programming Language (MSP430)
- Mastering Microcontroller: Timers, PWM, CAN, Low Power (MCU2)
- Learn FreeRTOS from Scratch

Languages

- Fluent in English, Hindi, Tulu, and Kannada